Supporting information for: Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

This supporting information accompanies the manuscript titled *Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling* and provides expanded workflow notes and supplemental derivations.

S1 Staged refinement workflow (expanded)

1. **Beam characterization (direct beam).** Acquire direct-beam images at several detector distances. Fit the spot position and shape versus distance to determine the incident-beam widths (w_x, w_z) and Gaussian divergences $(\Delta\theta_x, \Delta\theta_z)$ that set the instrument-limited broadening.

2. **Detector geometry calibration (3D powder standard).** Measure a polycrystalline hBN calibrant and subtract the detector dark frame. Fit multiple Debye–Scherrer rings in detector coordinates and refine detector distance, detector tilts (β, γ), in-plane detector rotation χ, and beam center (x_0, y_0).

3. **Sample alignment and goniometer misalignment.** For each dark-subtracted sample image, use indexed Bragg-peak centers rather than full-image intensity. With detector parameters fixed, refine sample alignment by minimizing detector-space residuals between measured and calculated peak positions spanning both specular and off-specular regions. Refine $(\theta_i, \delta, z_B, z_S)$ and the goniometer-axis misalignment rotations (α, ψ); include film dimensions (W, H) and thickness t to weight footprint and absorption.

4. **Mosaicity/profile refinement.** With geometry fixed, compare selected local detector ROIs or local $I(\phi)$ profiles after center alignment. The shared profile model is a wrapped Gaussian-plus-Lorentzian angular distribution, not a pseudo-Voigt common-width peak approximation. Off-specular arcs mainly constrain the Gaussian core, while off-condition specular intensity constrains the Lorentzian tail. Report Lorentzian FWHM, Gaussian FWHM, and the relative Lorentzian weight.

5. **Ordered-structure refinement.** With geometry and mosaicity fixed, refine the ordered intensity model against background-subtracted integrated counts in selected detector-space Bragg regions of interest (ROIs) from multiple dark-subtracted images. The corresponding prediction is the calculated detector intensity deposited in those same ROIs. Use one nuisance scale per image and physically constrained ordered-structure parameters.

6. **Disorder refinement.** After the ordered model converges, introduce stacking-disorder parameters only through fixed-Q_r rod profiles reported versus Q_z. The measured profile is formed from caked signal and normalization fields by summing signal and normalization over the selected rod mask before division. Geometry, detector calibration, beam terms, mosaicity, lattice constants, and the ordered baseline remain fixed; only stacking-disorder parameters and declared scale/background terms are refined.

All detector images are dark-subtracted before fitting. Any additional preprocessing (flat-field, polarization, solid-angle, or detector-efficiency corrections) should either be applied consistently to all images before refinement or stated explicitly if omitted. We recommend stage-specific uncertainty estimation by repeated constrained refits: resample the calibrant points for detector geometry, indexed peak lists for sample alignment, local reflection regions for mosaicity, accepted Bragg ROIs for ordered intensity, and selected rod bins for disorder. Report the standard deviation of the accepted solutions and propagate parameters to the next stage only after the preceding stage is stable.

S1.1 Coordinate remapping, projected profiles, and uncertainty propagation

The detector image remains the primary data object. The 2θ–ϕ representation is an overlap-weighted remapping used to evaluate fixed-m or fixed-Q_r trajectory masks and projected Q_z or L profiles. It is not treated as an independent measurement. The same remapping, masks, and profile extraction are applied to measured and calculated detector images.

The caked image is constructed with a sparse detector-to-cake operator

$$M \in \mathbb{R}^{B\times P}, \tag{S1}$$

where P is the number of detector pixels, B is the number of caked bins, and M_{bk} is the fractional overlap weight from detector pixel k into caked bin b. For detector signal s_k and normalization n_k,

$$S_b = \sum_k M_{bk} s_k, \qquad N_b = \sum_k M_{bk} n_k, \qquad I_b = \frac{S_b}{N_b}, \tag{S2}$$

for bins with $N_b > 0$. Signal and normalization are therefore accumulated before division. This prevents a projected profile from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

For each remapped bin, the calibrated detector and sample geometry define a scattering vector $\mathbf{Q}$. The coordinates used to label selected trajectories are

$$Q_r = |\mathbf{Q}_{\|}|, \qquad Q_z = \mathbf{Q}\cdot\hat{\mathbf{n}}, \qquad L = \frac{Q_z}{|\mathbf{c}^*|} = \frac{Q_z c}{2\pi}, \tag{S3}$$

where $\hat{\mathbf{n}}$ is the film normal. A selected fixed-Q_r or fixed-m family is retained when

$$|Q_r - Q_{r,0}| \le \Delta Q_r, \qquad Q_{z,\min} \le Q_z \le Q_{z,\max}. \tag{S4}$$

This region is simple in the physical Q_r–Q_z description but appears curved in the displayed 2θ–ϕ map because the coordinate transformation is nonlinear. The curved shape is therefore a consequence of the projection, not an additional physical model.

For a selected profile coordinate ℓ, let $r_{\ell b}$ be the mask or interpolation weight that assigns caked bin b to that profile point. The profile is

$$I_\ell^{\mathrm{ROI}} = \frac{\sum_b r_{\ell b} S_b}{\sum_b r_{\ell b} N_b} = \frac{\sum_k a_{\ell k} s_k}{\sum_k a_{\ell k} n_k}, \qquad a_{\ell k} = \sum_b r_{\ell b} M_{bk}. \tag{S5}$$

Each point in the plotted profile is therefore an average over a finite detector-space band. Its coordinate may be labeled by Q_z or L, but the point represents a resolution-limited projection rather than the intensity at one exact reciprocal-space coordinate.

The transformed-coordinate uncertainty has two parts: uncertainty in the coordinate label and uncertainty in the rebinned intensity. The coordinate-label uncertainty follows from the finite detector-pixel footprint and from uncertainty in the detector calibration. For square detector pixels with pitch p, detector distance d, and $t = 2\theta$, the leading flat-detector terms are

$$\sigma_{2\theta,\mathrm{pix}}(t) = \frac{p\cos^2 t}{\sqrt{12}\, d}, \qquad \sigma_{\phi,\mathrm{pix}}(t) = \frac{p}{\sqrt{12}\, d \tan t}. \tag{S6}$$

The radial term is finite and largest near the direct beam. The azimuthal term scales as $1/\tan(2\theta)$, so the problematic regime is confined to the near-specular, low-2θ region where azimuth is poorly conditioned.

For the $d = 75$ mm and $p = 300\,\mu$m geometry used in the uncertainty calculation, the maximum radial coordinate uncertainty is approximately $0.066°$. The azimuthal coordinate uncertainty falls below $1°$ above $2\theta \approx 3.8°$, below $0.5°$ above $2\theta \approx 7.5°$, and below $0.25°$ above $2\theta \approx 14.9°$. More generally, the threshold for a target azimuthal width w_ϕ is

$$2\theta_{\mathrm{crit}} = \arctan\left(\frac{p}{\sqrt{12}\, d\, w_\phi}\right), \tag{S7}$$

with w_ϕ expressed in radians.

Additional calibration uncertainty can be propagated with the usual Jacobian form. If the uncertain geometry parameters are collected in $\mathbf{g}$ with covariance matrix Σ_g, then

$$\sigma_{2\theta,\mathrm{cal}}^2 = J_{2\theta} \Sigma_g J_{2\theta}^{\mathsf{T}}, \qquad \sigma_{\phi,\mathrm{cal}}^2 = J_\phi \Sigma_g J_\phi^{\mathsf{T}}. \tag{S8}$$

If the displayed bin-center label is also treated as uncertain, the finite bin width contributes

$$\sigma_{2\theta,\mathrm{bin}}^2 = \frac{\Delta(2\theta)^2}{12}, \qquad \sigma_{\phi,\mathrm{bin}}^2 = \frac{\Delta\phi^2}{12}. \tag{S9}$$

These coordinate terms describe the uncertainty of the transformed axis value. They are separate from intensity uncertainty.

The intensity uncertainty is inherited from the detector through the same overlap weights used to construct the caked image and the projected profile. If v_k is the detector-pixel variance and the detector-pixel variances are independent, then

$$\mathrm{Var}\left(I_\ell^{\mathrm{ROI}}\right) = \frac{\sum_k a_{\ell k}^2 v_k}{\left(\sum_k a_{\ell k} n_k\right)^2}. \tag{S10}$$

If overlapping interpolation weights are used, neighboring profile points can be statistically correlated. The corresponding covariance is

$$\mathrm{Cov}\left(I_\ell^{\mathrm{ROI}}, I_q^{\mathrm{ROI}}\right) = \frac{\sum_k a_{\ell k} a_{qk} v_k}{\left(\sum_k a_{\ell k} n_k\right)\left(\sum_k a_{qk} n_k\right)}. \tag{S11}$$

Thus the same weights that define the projected profile also propagate the detector-pixel variance. The covariance between neighboring projected points is neglected in the present line-shape comparison unless formal profile error bars are required.

The displayed coordinate for a projected point can be represented by the same finite-band weighting. For example,

$$\bar{L}_\ell = \frac{\sum_b r_{\ell b} N_b L_b}{\sum_b r_{\ell b} N_b}, \qquad \sigma^2_{L,\ell} = \frac{\sum_b r_{\ell b} N_b (L_b - \bar{L}_\ell)^2}{\sum_b r_{\ell b} N_b}, \tag{S12}$$

before adding calibration, incidence-angle, wavelength, divergence, and mosaic contributions. This width is a projection and resolution effect. It is not a counting-statistical error bar.

These projection uncertainties are coupled to model-parameter correlations. Peak positions are sensitive to detector distance, beam center, detector tilt, sample height, and incidence angle. Profile widths are sensitive to both beam divergence and the Gaussian mosaic core. Weak off-condition and low-L intensities are sensitive to wavelength bandwidth, background, and the Lorentzian mosaic tail. Relative integrated intensities are sensitive to scale, footprint, absorption, thickness, and structure-factor parameters. The staged workflow separates these effects by fixing or constraining the parameters tied to peak position before fitting profile widths, and by fixing the geometry and mosaic state before interpreting ordered intensities.

The practical consequences are: (i) the 2θ–ϕ and Q_r/Q_z profiles are controlled projections of the detector image, not independent ideal reciprocal-space scans; (ii) transformed-coordinate uncertainty is concentrated near the low-angle/specular region; (iii) intensity variance is propagated through the same overlap weights used for caking and profile extraction; and (iv) the strongest remaining systematic coupling occurs near low L, where direct-beam background, bandwidth, divergence, incidence angle, and the Lorentzian mosaic tail contribute to the same detector-space feature.

S1.2 Wrapped mosaic distribution

In the main text we describe out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$ in the crystallite tilt angle $\Delta\theta$ relative to the substrate normal. Since $\Delta\theta$ is an orientation angle, $\Delta\theta$ and $\Delta\theta + 2\pi$ represent the same physical state. Accordingly, ω is treated as a 2π-periodic probability density on $\Delta\theta \in (-\pi, \pi]$.

The implemented profile is a wrapped Gaussian plus a wrapped Lorentzian,

$$\omega_{\mathrm{w}}(\Delta\theta; \eta, \Gamma, \sigma) = \eta\, L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1-\eta)\, G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \le \eta \le 1.$$

This is not a pseudo-Voigt constraint: the Lorentzian half-width Γ and Gaussian standard deviation σ are independent. The wrapped Lorentzian and wrapped Gaussian are

$$L_{\mathrm{w}}(\Delta\theta; \Gamma) = \sum_{m=-\infty}^{\infty} \frac{1}{\pi} \frac{\Gamma}{(\Delta\theta + 2\pi m)^2 + \Gamma^2} = \frac{1}{2\pi} \frac{\sinh \Gamma}{\cosh \Gamma - \cos \Delta\theta},$$

$$\begin{aligned} G_{\mathrm{w}}(\Delta\theta; \sigma) &= \sum_{m=-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta\theta + 2\pi m)^2}{2\sigma^2} \right] \\ &= \frac{1}{2\pi} \left[1 + 2 \sum_{n=1}^{\infty} e^{-n^2\sigma^2/2} \cos(n\Delta\theta) \right]. \end{aligned}$$

Both components integrate to unity over $(-\pi, \pi]$, so ω_{w} is normalized. The reported Lorentzian FWHM is 2Γ, and the reported Gaussian FWHM is $2\sqrt{2\ln 2}\,\sigma$. The local small-width limit recovers the usual line-shape widths, but the refinement and normalization are defined by the wrapped functions above.

For a reflection with Bragg-sphere radius $G \equiv |\mathbf{G}|$, mosaicity redistributes intensity in polar angle ϑ with line density $I_0\,\omega_{\mathrm{w}}(\vartheta - \vartheta_B)\,d\vartheta$, where ϑ_B is the Bragg position for that reflection. Random in-plane rotations about the film normal revolve this line density about the G_z axis, yielding an axially symmetric surface density $\rho(\vartheta)$. With area element $dA = 2\pi G^2 \sin\vartheta\,d\vartheta$, conservation of intensity gives

$$\rho(\vartheta) = \frac{I_0}{2\pi G^2 \sin\vartheta}\,\omega_{\mathrm{w}}(\vartheta - \vartheta_B), \qquad 0 < \vartheta < \pi,$$

as used in the main text.

S1.3 Wave propagation, refraction, and absorption

Wave attenuation occurs both at the interfaces (Fresnel transmission) and as the wave propagates through the film (absorption). In the forward model we therefore (i) refract each ray into and out of the film, (ii) assign the absorption to the out-of-plane component of the internal wavevector, and (iii) weight scattering from a finite thickness t by thickness-averaged Beer–Lambert attenuation. This subsection records the explicit equations used.

Complex refractive index. We write the complex refractive index as

$$\tilde{n} = 1 - \delta_n + i\beta_n, \tag{S13}$$

where the subscript n avoids confusion with the geometric angles (δ, β) used elsewhere. In terms of the wavelength λ, electron density ρ_e, classical electron radius r_e, and linear absorption coefficient μ,

$$\delta_n = \frac{r_e \lambda^2 \rho_e}{2\pi}, \qquad \beta_n = \frac{\mu\lambda}{4\pi}. \tag{S14}$$

Angles and refraction. Let $k_0 = 2\pi/\lambda$ and take $\hat{z}$ along the sample normal. For an incident ray in air, the grazing-incidence angle θ_i and in-plane azimuth φ_i may be expressed in terms of the external wavevector components as

$$\sin\theta_i = \frac{k_{i,z}}{k_0}, \tag{S15}$$

$$\tan\varphi_i = \frac{k_{i,x}}{k_{i,y}}. \tag{S16}$$

Upon refraction, we compute the transmitted grazing angle using the real part of the refractive index,

$$\theta_t = \cos^{-1}\left(\frac{\cos\theta_i}{\mathrm{Re}(\tilde{n})}\right). \tag{S17}$$

Internal wavevector components. We define a real internal wavevector magnitude for the in-plane components as

$$k' = k_0 \left(\mathrm{Re}\left(\tilde{n}^2\right)\right)^{1/2}, \tag{S18}$$

so that

$$k_{tx} = k' \cos\theta_t \sin\varphi_i, \tag{S19}$$

$$k_{ty} = k' \cos\theta_t \cos\varphi_i. \tag{S20}$$

To satisfy the boundary condition at the surface, we take only the out-of-plane component to be complex and write

$$k_{tz}^2 = a + ib, \qquad k_{tz} = u + iv = \sqrt{a + ib}. \tag{S21}$$

Expanding $(u + iv)^2$ gives

$$(u + iv)^2 = u^2 - v^2 + 2iuv = a + ib,$$

so that the real and imaginary parts satisfy

$$a = u^2 - v^2 = (u + v)(u - v), \qquad b = 2uv = (u + v)^2 - u^2 - v^2. \tag{S22}$$

The magnitude follows from

$$|k_{tz}|^2 = u^2 + v^2 = \sqrt{(a + ib)(a - ib)} = \sqrt{a^2 + b^2}, \tag{S23}$$

which yields

$$u^2 = \frac{\sqrt{a^2 + b^2} + a}{2}, \qquad v^2 = \frac{\sqrt{a^2 + b^2} - a}{2}. \tag{S24}$$

In our parameterization,

$$\mathrm{Re}(k_{tz}^2) = k'^2 \sin^2\theta_t \equiv a_t, \tag{S25}$$

$$\mathrm{Im}(k_{tz}^2) = k_0^2\ \mathrm{Im}\left(\tilde{n}^2\right) \equiv b, \tag{S26}$$

so that

$$\mathrm{Re}(k_{tz}) = \sqrt{\frac{\sqrt{a_t^2 + b^2} + a_t}{2}}, \tag{S27}$$

$$\mathrm{Im}(k_{tz}) = \sqrt{\frac{\sqrt{a_t^2 + b^2} - a_t}{2}}. \tag{S28}$$

We take the physical branch $\mathrm{Im}(k_{tz}) > 0$ so that the field decays into the film.

Exit path and refraction out of the film. For the scattered ray we repeat the same construction using the external grazing-exit angle and in-plane azimuth (θ_f, φ_f). We compute an internal exit angle θ_t' (again using $\mathrm{Re}(\tilde{n})$) and form

$$k_{tx}' = k' \cos\theta_t' \sin\varphi_f, \tag{S29}$$

$$k_{ty}' = k' \cos\theta_t' \cos\varphi_f, \tag{S30}$$

$$a_f \equiv k'^2 \sin^2\theta_t', \tag{S31}$$

with the same $b_f = b_t = b$,

$$\mathrm{Re}(k'_{tz}) = \sqrt{\frac{\sqrt{a_f^2 + b^2} + a_f}{2}}, \tag{S32}$$

$$\mathrm{Im}(k'_{tz}) = \sqrt{\frac{\sqrt{a_f^2 + b^2} - a_f}{2}}. \tag{S33}$$

Optical reversibility then gives the refraction back into air; inverting the relation for θ_t gives

$$\theta_f = \cos^{-1}(\mathrm{Re}(\tilde{n}) \cos \theta'_t) \,. \tag{S34}$$

Fresnel transmission factors. We weight each ray by Fresnel transmission factors at entry and exit interfaces. For a scalar field written in terms of the normal wavevector components, one convenient form for an amplitude transmission coefficient from medium 1 to 2 is

$$t_{12} = \frac{2k_{1z}}{k_{1z} + k_{2z}}. \tag{S35}$$

Absorption weight (uniform scattering through thickness). Assuming a uniform probability of scattering across depth $z \in [0, t]$ and Beer–Lambert attenuation along the incident and exit paths, the intensity weight used in the forward model is

$$w_{\mathrm{abs}} = \frac{1}{t} \int_0^t \exp\left[-\mu z \left(\frac{1}{\sin\theta_i} + \frac{1}{\sin\theta_f}\right)\right] dz \tag{S36}$$

$$= \frac{1 - \exp\left[-\mu t \left(\frac{1}{\sin\theta_i} + \frac{1}{\sin\theta_f}\right)\right]}{\mu t \left(\frac{1}{\sin\theta_i} + \frac{1}{\sin\theta_f}\right)}. \tag{S37}$$

S1.4 Detector-derived ordered and disorder observables

The detector-space forward model predicts full images, but the inverse problems use stage-specific observables. In the ordered-structure stage, the measured quantity for ROI i in image m is the background-subtracted detector count

$$y_{mi} = \sum_{(u,v)\in\Omega_{mi}} \left[I_m^{\mathrm{meas}}(u,v) - b_{mi}(u,v)\right],$$

where Ω_{mi} is the selected detector-space Bragg ROI and b_{mi} is the local background model. The corresponding prediction is the calculated detector intensity deposited in the same ROI,

$$\mu_{mi} = \sum_{j\in\mathcal{R}_{mi}} w_j,$$

with geometry, beam terms, and mosaicity fixed. One nuisance scale per image accounts for exposure and flux differences. This makes the ordered stage a multi-image ROI-integrated Bragg-intensity refinement, not a full-image pixel objective.

The PbI_2 disorder refinement is a final-stage selected-profile fit, not a second global image refinement. The upstream beam, detector, sample alignment, mosaic/profile parameters, lattice

constants, and ordered PbI_2 baseline are fixed before this stage starts. The detector image is first caked into a signal field $S(\phi, 2\theta)$ and a normalization field $N(\phi, 2\theta)$ using the calibrated detector geometry. For the selected-rod conversion, let η denote the polar angle of the scattering vector relative to the film normal. For each caked bin,

$$Q = \frac{4\pi}{\lambda} \sin\left(\frac{2\theta}{2}\right), \qquad Q_r = |\sin\eta| Q, \qquad Q_z = \cos\eta\, Q.$$

A selected rod with in-plane radius $Q_{r,0}$ is retained when

$$|Q_r - Q_{r,0}| \le \Delta Q_r, \qquad Q_{z,\min} \le Q_z \le Q_{z,\max}.$$

A branch selector may be applied after this reciprocal-space mask so that one branch is used for the default fit and the symmetry-related branch is retained for validation.

Let S_{ij} and N_{ij} denote the caked signal and normalization fields, and let w_{ij} denote the selected-rod mask. The measured profile is

$$I_j^{\mathrm{rod}} = \frac{\sum_i w_{ij} S_{ij}}{\sum_i w_{ij} N_{ij}},$$

for bins with positive selected normalization. Signal and normalization are therefore summed before division. This prevents the final trace from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

The model profile is evaluated along the same selected branch. In compact notation,

$$M_j(\boldsymbol{\theta}_{\mathrm{dis}}) = B(x_j) + s\,[K * I_m^{\mathrm{nc}}(L; \boldsymbol{\theta}_{\mathrm{dis}})]_j,$$

where x_j is the profile coordinate, L is the corresponding stacking coordinate on the selected branch, B is a low-order background, s is a global scale, K is the fixed detector/profile broadening kernel, and I_m^{nc} is the no-correlation analytical stacking intensity from Sec. S3. When an explicit closed-form broadening kernel is used, K is written as a Gaussian-plus-Lorentzian kernel with widths fixed by the upstream resolution and mosaic/profile state or declared before the disorder fit. It is not used to release independent peak areas.

The disorder objective is evaluated only over the retained profile bins,

$$\chi^2_{\mathrm{dis}}(\boldsymbol{\theta}_{\mathrm{dis}}, s, B) = \sum_{j \in \mathcal{J}} \rho\left(\frac{I_j^{\mathrm{rod}} - M_j(\boldsymbol{\theta}_{\mathrm{dis}})}{\sigma_j}\right),$$

where ρ may be a least-squares or robust loss and $\mathcal{J}$ is the retained Q_z interval. The known Bragg locations on the selected branch are inherited from the fixed detector geometry and ordered model. Peak amplitudes are not fit independently before the stacking model is applied; otherwise the diffuse between-peak intensity would already have been absorbed by the preprocessing.

S2 Material background and stacking–fault model

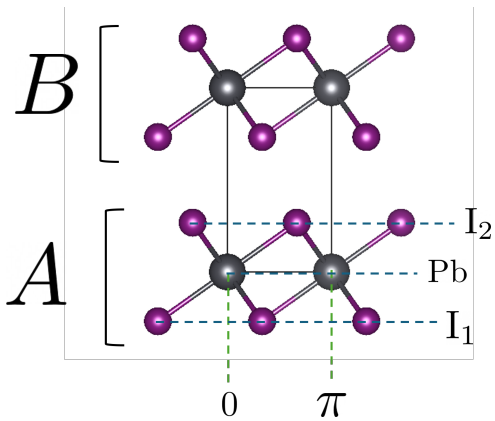
In layered metal–halides such as PbI_2 each I–Pb–I sandwich behaves like a rigid trilayer that can shift laterally with negligible energy cost compared with the much stronger intra-layer bonding. Consequently the crystal can adopt several polytypes that differ only in the stacking sequence of these sandwiches along the c axis. Here we focus on the two sequences that bracket the experimentally relevant window:

S2.1 Polytypism of PbI_2

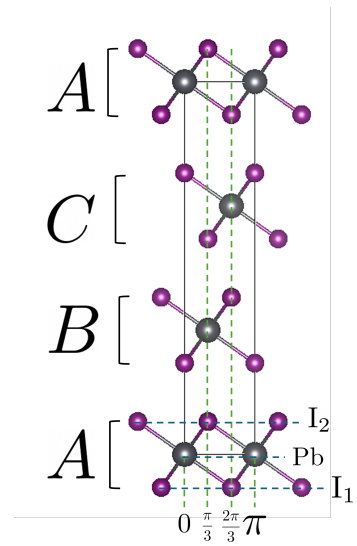
2H ($P\bar{3}$m1) AB repeat stable below 140 °C; no lateral shift ($\phi_{AB} = 0$); see Fig. S1a.

6H ($R\bar{3}$) ABCA repeat above ≈ 200 °C; lateral shift $a/\sqrt{3}$ gives $\phi = 2\pi/3$; 2H ⇔ 6H transition near 94 °C; see Fig. S1b.

Our samples never achieve perfect periodicity; stacks often contain finite 2H and 6H "domains" interspersed by stacking faults. To model the resulting diffuse scattering we adapt the Hendricks–Teller (HT) Markov formalism and follow the streamlined matrix notation introduced a few years later by Lifshitz, who showed how to re-express pair probabilities through the eigenvalues of a 3×3 transition matrix.



(a) 2H polytype (AB) with no in-plane phase shifts.



(b) 6H R$\bar{3}$ polytype (ABCA) with $\frac{1}{3}$-lattice translations giving $e^{i2\pi/3}$ phase steps.

Figure S1: Low-temperature 2H and high-temperature 6H stacking sequences in PbI_2. Arrows illustrate the lateral shifts responsible for the phase factors used in the HT–Lifshitz model.

S2.2 Molecular structure factor for PbI_2

A single layer form factor is

$$F_m = b_{Pb} + b_I \left[e^{2\pi i(H/3+2K/3+Lz/3)} + e^{2\pi i(2H/3+K/3-Lz/3)} \right] \tag{S38}$$

Where now the amplitude will be

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \tag{S39}$$

Where $\vec{R_p} = m\vec{a} + n\vec{b}$ is the in-plane position vector, with $m, n \in \mathbb{Z}$. Then $\vec{Q}\cdot\vec{R_p} \to Hm + kn$, and the Bragg condition makes H, K integers.

S3 No-correlation stacking-fault model

Let $\tau_j \in \{0,1\}$ indicate whether the j-th interface slips ($\tau_j = 1$) or not ($\tau_j = 0$). The structure factor of one layer is $F_0(h,k,\ell)$, and for a three-layer (ABC) repeat the per-layer phase advance is

$$\phi(h,k,\ell) = \frac{h+2k}{3} + \ell/3, \tag{S40}$$

with h, k, ℓ in reciprocal-lattice units.
We now add the phase of each layer to the form factor from equation S39.

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \sum_{\ell} e^{i\phi(h,k,\ell)\ell} \tag{S41}$$

Writing $e^{i\phi\ell} = \prod_{j=1}^{\ell} e^{i\phi}$

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R_p}} \prod_{\ell} e^{i\phi(h,k,\ell)\ell} \tag{S42}$$

S3.1 Coherent amplitude of an N-layer stack

At layer ℓ the factor multiplying F_0 is

$$\left[1 - \tau_\ell + \tau_\ell e^{-i\phi}\right] e^{i\phi\ell},$$

so that

$$A = F_0 \sum_{\ell=0}^{N-1} \left[\prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi})\right] e^{i\phi\ell}. \tag{S43}$$

S3.2 Uncorrelated faults

Assume the τ_j are i.i.d. with slip probability $p = \Pr(\tau_j = 1)$. Independence gives

$$\left\langle \prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi}) \right\rangle = \left[(1-p) + p\, e^{-i\phi}\right]^{\ell} \equiv f^{\,\ell}, \qquad f \equiv (1-p) + p\, e^{-i\phi}. \tag{S44}$$

S3.3 Mean amplitude

Averaging (S43) and inserting (S44) gives the geometric series

$$\langle A \rangle = F_0 \sum_{\ell=0}^{N-1} f^{\ell} e^{i\phi\ell} = F_0 \frac{1 - (f\, e^{i\phi})^N}{1 - f\, e^{i\phi}}. \tag{S45}$$

S3.4 Infinite-crystal limit

For any non-zero fault probability ($|f| < 1$),

$$\boxed{\lim_{N\to\infty} \langle A \rangle = F_0 \frac{f}{1 - f\, e^{i\phi}}} \tag{S46}$$

The perfect-crystal result is recovered as $f \to 1$.

S3.5 Mean-square amplitude

The heights of sharp Bragg peaks: i.e. the square of the coherently averaged structure amplitude. This is only the part of the intensity that survives perfect phase coherence: sharp Bragg peaks. Any layer-to-layer fluctuations are averaged before squaring and thus disappear.

$$
\begin{aligned}
|\langle A\rangle|^2 &= |F_0|^2 \frac{f^2}{|1 - f\, e^{i\phi}|^2} \\
&= |F_0|^2 \frac{f^2}{(1 - f\cos\phi)^2 + f^2 \sin^2\phi} && \text{(S47)} \\
&= |F_0|^2 \frac{f^2}{(1-f)^2 + 4f\sin^2\dfrac{\phi}{2}}. && \text{(S48)}
\end{aligned}
$$

S3.6 Ensemble-averaged intensity

The full detector signal; adds the broadened peaks and diffuse background arising from stacking faults.

$$< |A|^2 > \tag{S49}$$

S3.6.1 Layer Autocorrelation

The layer-resolved autocorrelation of the stacking sequence tells you how phase coherence decays with layer phase shifts. It is the kernel that converts local disorder into the full intensity giving both Bragg and diffuse components.

Define

$$g_j \equiv 1 - \tau_j + \tau_j e^{-i\phi}, \qquad S_\ell \equiv \prod_{j=0}^{\ell} g_j. \tag{S50}$$

We wish to compute

$$\langle S_\ell S_{\ell'}^* \rangle = \langle \prod_{j=0}^{\ell} g_j \prod_{j'=0}^{\ell'} g_{j'}^* \rangle. \tag{S51}$$

Since the g_j are i.i.d., expectations of products factor, and $g_j g_j^* = 1$ identically. We split into two cases:

Case A: $\ell \geq \ell'$.

$$S_\ell S_{\ell'}^* \;=\; \underbrace{\Big(\prod_{j=0}^{\ell'} g_j\Big)\Big(\prod_{j'=0}^{\ell'} g_{j'}^*\Big)}_{=\,1\ \text{(common layers)}} \;\underbrace{\Big(\prod_{j=\ell'+1}^{\ell} g_j\Big)}_{\text{extra layers}} \tag{S52}$$

- For $0 \leq j \leq \ell'$, $g_j g_j^* = 1$, so the first two products give 1. - For $j = \ell'+1, \ldots, \ell$, independence implies

$$\Big\langle \prod_{j=\ell'+1}^{\ell} g_j \Big\rangle = \prod_{j=\ell'+1}^{\ell} \langle g_j \rangle = f^{\,\ell-\ell'}. \tag{S53}$$

Hence

$$\langle S_\ell S_{\ell'}^* \rangle = f^{\,\ell-\ell'}, \qquad (\ell \geq \ell'). \tag{S54}$$

Case B: $\ell < \ell'$. By an identical argument, swapping roles,

$$\langle S_\ell S_{\ell'}^* \rangle = (f^*)^{\ell'-\ell}, \qquad (\ell < \ell').$$

The Solutions are then

$$\boxed{\langle S_\ell S_{\ell'}^* \rangle = \begin{cases} f^{\,\ell-\ell'}, & \ell \geq \ell', \\ (f^*)^{\ell'-\ell}, & \ell < \ell'. \end{cases}} \tag{S55}$$

S3.7 Single-sum form of the double series

With $\ell'' = \ell - \ell'$, the double sum becomes

$$\sum_{\ell=0}^{N_z-1} \sum_{\ell'=0}^{N_z-1} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \sum_{\ell''=-N_z+1}^{N_z-1} f^{|\ell''|} e^{i(\phi-\psi)\ell''}. \tag{S56}$$

Splitting out $\ell'' = 0$ and the $\pm\ell''$ pairs yields

$$N_z \left[1 + \sum_{\ell''=1}^{N_z-1} f^{\ell''} \left(e^{i(\phi-\psi)\ell''} + e^{-i(\phi-\psi)\ell''} \right) \right],$$

with

$$\sum_{\ell''=1}^{N_z-1} f^{\ell''} e^{\pm i(\phi-\psi)\ell''} = \frac{f\, e^{\pm i(\phi-\psi)} \left[1 - f^{N_z-1} e^{\pm i(\phi-\psi)(N_z-1)} \right]}{1 - f\, e^{\pm i(\phi-\psi)}}.$$

S3.8 Infinite-crystal limit $(0 \leq f < 1)$

Letting $N_z \to \infty$ gives

$$\sum_{\ell,\ell'} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \, \frac{1 - f^2}{1 + f^2 - 2f\cos(\phi - \psi)}. \tag{S57}$$

S3.9 Scattering cross-section

Substituting (S57),

$$\boxed{\frac{d\sigma}{d\Omega} = |F_m|^2 \, \frac{(2\pi)^2}{A_c} \, \delta(\mathbf{Q} - \mathbf{G}_p) \, N_p N_z \, \frac{1 - f^2}{1 + f^2 - 2f\cos(\phi - \psi)}}. \tag{S58}$$

This is identical to the Hendrix–Teller formula upon identifying $\phi^{\mathrm{HT}} = \phi - \psi$ and noting that their parameter $\cos(\phi^1 - \phi^2)$ equals f.